

Online Supplement to “Confinement-loss prediction in photonic crystal fiber SPR sensors: kernel surrogates versus generative data augmentation”

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Abstract

This supplement contains material supporting the main article that is not required to follow its argument: the dual formulation of support vector regression and its optimality conditions (Sect. S1), the Bayesian hyperparameter search space and its outcome (Sect. S2), per-fold leave-one-geometry-out results (Sect. S3), the generative model configuration and the contradiction-search protocol (Sect. S4), a Gaussian process kernel and regulariser ablation (Sect. S5), and the complete Monte Carlo tolerance tables (Sect. S6). Section, table, figure and equation numbers are prefixed with S; citations refer to the reference list of the main article, reproduced here for convenience.

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S1 Support vector regression: dual formulation

The main article states the primal problem, Eqs. (3)–(4). Introducing Lagrange multipliers $\alpha_i, \alpha_i^* \geq 0$ for the two ϵ -tube constraints and $\eta_i, \eta_i^* \geq 0$ for the non-negativity of the slacks gives the Lagrangian

$$\begin{aligned}
L = & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) - \sum_{i=1}^n (\eta_i \xi_i + \eta_i^* \xi_i^*) \\
& - \sum_{i=1}^n \alpha_i (\epsilon + \xi_i - y_i + \langle w, x_i \rangle + b) \\
& - \sum_{i=1}^n \alpha_i^* (\epsilon + \xi_i^* + y_i - \langle w, x_i \rangle - b). \tag{S1}
\end{aligned}$$

Setting the partial derivatives with respect to the primal variables to zero yields the saddle-point conditions

$$w = \sum_{i=1}^n (\alpha_i - \alpha_i^*) x_i, \quad \sum_{i=1}^n (\alpha_i - \alpha_i^*) = 0, \quad \alpha_i^{(*)} + \eta_i^{(*)} = C, \tag{S2}$$

and substituting them back removes the primal variables entirely, leaving the dual

$$\begin{aligned}
\max_{\alpha, \alpha^*} \quad & -\frac{1}{2} \sum_{i,j=1}^n (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*) K(x_i, x_j) \\
& - \epsilon \sum_{i=1}^n (\alpha_i + \alpha_i^*) + \sum_{i=1}^n y_i (\alpha_i - \alpha_i^*), \tag{S3}
\end{aligned}$$

subject to $\sum_i (\alpha_i - \alpha_i^*) = 0$ and $\alpha_i, \alpha_i^* \in [0, C]$. The prediction is then

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) K(x_i, x) + b. \tag{S4}$$

Two properties of (S3)–(S4) matter for the argument in the main article. First, the inputs enter only through the kernel $K(x_i, x_j)$, so the RBF map never has to be formed explicitly. Second, the Karush-Kuhn-Tucker complementarity conditions

$$\alpha_i (\epsilon + \xi_i - y_i + \langle w, x_i \rangle + b) = 0, \quad (C - \alpha_i) \xi_i = 0, \tag{S5}$$

force $\alpha_i = \alpha_i^* = 0$ for every sample strictly inside the ϵ -tube. The expansion in (S4) is therefore supported only by samples on or outside the tube boundary. On this dataset that concentrates the fit near the resonance transition, where the response is steepest, and it is why the model tolerates a training set of 432 samples.

Condition (S5) also explains the contrast with Gaussian process regression in Sect. 5.2 of the main article. A pair of near-duplicate inputs carrying contradictory targets simply saturates at $\alpha_i = C$: both points become bounded support vectors, the fitted function is worse, but nothing in (S3) becomes ill-posed. There is no matrix to invert.

S2 Bayesian optimisation of the SVR hyperparameters

Table S1 gives the search space and the selected configuration. The acquisition function is expected improvement over a Gaussian process surrogate on the hyperparameter space, with 32 evaluations. Each candidate is scored by three-fold cross-validated mean squared error computed on the training partition only, so no information from the held-out geometry enters the selection.

Table S1 Hyperparameter search space and the configuration selected by Bayesian optimisation. Priors are log-uniform because the three parameters span several orders of magnitude and the objective is far more sensitive to their order than to their mantissa.

Parameter	Range	Prior	Selected
C (box constraint)	$[1, 2000]$	log-uniform	2000
γ (RBF width)	$[10^{-4}, 1]$	log-uniform	0.0172
ϵ (tube)	$[10^{-4}, 10^{-1}]$	log-uniform	10^{-4}
Kernel	RBF (fixed)	—	RBF
Evaluations	32	—	—
Inner validation	3-fold CV	—	—

The selected C and ϵ sit at their respective range limits, which indicates the optimum lies at or beyond the boundary: the fit prefers a hard penalty and an almost vanishing insensitivity tube. Widening the range is unlikely to change the fit materially – at $\epsilon = 10^{-4}$ the tube is already far narrower than the residual scale – but the boundary solution should be reported rather than presented as an interior optimum.

The selected width $\gamma = 0.0172$ is small relative to the unit-cube extent of the scaled inputs, giving a broad kernel: the model is smoothing over a substantial fraction of the design space rather than fitting locally, which is the expected regime when nine geometries must support prediction on a tenth.

S3 Per-fold leave-one-geometry-out results

Table S2 gives the fold-level results summarised in Table 2 of the main article, and Fig. S1 shows them graphically.

The per-fold view supports a claim the averages cannot. Augmentation is not a uniform penalty: it is nearly neutral on D3 and D4, mildly helpful on D5, and catastrophic on D8, where the error grows by a factor of 23. This is the signature of synthetic samples that contradict specific regions of the design space rather than of noise added evenly across it. D5 is the only fold where augmentation helps, and it is also the fold with the worst real-only error – consistent with extra samples helping only where the real coverage was genuinely thin.

Table S2 Held-out mean squared error for each of the nine geometric configurations, with and without GAN-generated training samples. Folds run down the rows; the final two rows give the summary statistics quoted in the main article.

Held-out geometry	SVR, real data only	SVR + GAN samples	Ratio
D1	0.128	1.074	8.41
D2	0.379	0.848	2.24
D3	0.257	0.321	1.25
D4	0.212	0.261	1.23
D5	0.829	0.487	0.59
D6	0.121	1.359	11.27
D7	0.131	0.684	5.21
D8	0.096	2.200	22.83
D9	0.060	0.194	3.24
Mean	0.246	0.825	—
Standard deviation	0.226	0.610	—

Ratio is augmented MSE divided by real-only MSE; values above one indicate augmentation made that fold worse.

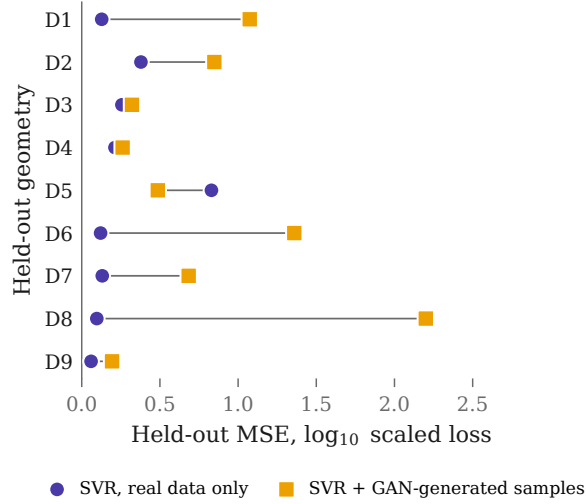


Fig. S1 Per-fold held-out error with and without GAN augmentation. Each connector spans one geometry, so the direction of change is read directly. Eight of the nine folds move to the right, that is, get worse under augmentation.

S4 Generative model and contradiction search

S4.1 WGAN-GP configuration

The generative model reimplements the architecture reported by Zelaci et al. [1], so that the augmentation experiment tests their procedure and not a variant of it. Table S3 gives the configuration.

Table S3 WGAN-GP configuration used for the augmentation experiments. Generator and critic are given side by side; the training column applies to both.

Property	Generator	Critic	Training
Hidden layers	5	5	—
Units per layer	128	32	—
Activation	LeakyReLU (0.2)	LeakyReLU (0.2)	—
Normalisation	none	batch normalisation	—
Output width	8	1	—
Optimiser	—	—	Adam, $\beta = (0.5, 0.9)$
Learning rate	—	—	10^{-4}
Batch size	—	—	32
Critic steps	—	—	5 per generator step
Epochs	—	—	2000

The generator is trained on the concatenation of the min-max scaled input features and the min-max scaled target, so it models the joint distribution of geometry and response. This is the property the contradiction search exploits: nothing in the objective requires the generated joint samples to be consistent with a single-valued map from geometry to loss.

S4.2 Contradiction search protocol

We drew 50,000 samples from the trained generator and, for each, found its nearest neighbour in the scaled feature space under the Euclidean metric, excluding the sample itself. For a pair at feature distance δ with targets differing by Δy , we ranked by

$$\rho = \frac{\Delta y}{\max(\delta, 10^{-6})}, \quad (\text{S6})$$

which measures how steeply the generated response surface must rise to accommodate the pair. Table 4 of the main article reports the three pairs with the largest ρ . Ranking by ρ rather than by δ alone is deliberate: a pair that is merely close is unremarkable, and it is the combination of proximity and target disagreement that is physically impossible.

The feature space for this search is the seven-dimensional set the generator was trained on – analyte index, real effective index, wavelength, pitch and the three hole diameters – min-max scaled to the unit cube. The floor of 10^{-6} in (S6) guards the division for coincident samples.

Note that ρ is a diagnostic, not an estimator: its absolute magnitude depends on the scaling of both axes and should not be compared across datasets. Within this experiment it is the ordering that carries the information.

S5 Gaussian process kernel and regulariser ablation

Section 5.2 of the main article attributes the GPR collapse to ill-conditioning of $[K + \sigma_n^2 I]$. If that account is right, the collapse should depend on how the inversion is

regularised. Table S4 tests this by crossing the two available regularisers – an additive white-noise kernel term (WK) and `scikit-learn`’s explicit jitter α – against real and augmented training data.

Table S4 GPR held-out MSE under each combination of regulariser, on real and on augmented training data. Regulariser configurations run down the rows, training sets across the columns.

Regulariser configuration	Real data	GAN-augmented
White kernel only ($\alpha = 0$)	0.185	69.873
White kernel and α	2.116	68.262
α only, no white kernel	71.849	69.873
Neither	—	—

Dashes mark configurations for which no value was recorded. With neither regulariser present, the covariance matrix is singular by construction once two training inputs coincide, so the configuration is not expected to produce a finite result.

On real data the regulariser choice matters enormously: the white kernel alone gives 0.185, while jitter alone gives 71.849 – a 388-fold difference produced by nothing but how the diagonal is stabilised. This is already evidence that the inversion is close to the edge of conditioning even before any synthetic data is introduced.

On augmented data the picture inverts: every configuration lands near 69, regardless of regulariser. Once the contradictory pairs of Sect. S4 are present, no choice of regularisation recovers a usable model, because the problem is no longer conditioning of an otherwise well-posed system but a training set that asks for two different outputs at effectively the same input. The regulariser can absorb noise; it cannot absorb contradiction.

S6 Monte Carlo tolerance analysis

S6.1 Protocol

The baseline configuration is the nominal design of Table 1 of the main article at analyte index 1.33. For each tolerance level $\sigma \in \{1\%, 3\%, 5\%\}$ we drew 100 perturbed geometries, adding to each geometric variable v a Gaussian perturbation of standard deviation σv , and located the predicted resonance by taking the argmax of the surrogate prediction over a 500-point wavelength grid spanning the dataset range. The reported jitter is the standard deviation of the resulting peak positions.

Per-variable volatilities were obtained by repeating the procedure at $\sigma = 3\%$ with one variable perturbed at a time and the rest held at nominal.

S6.2 Results

Table S5 reports both panels. The upper panel gives the scatter when all four geometric variables are perturbed together, and the lower panel attributes that scatter to individual variables.

Table S5 Monte Carlo peak scatter at each tolerance level, and the per-variable decomposition at 3%. Both panels report the standard deviation of the predicted resonance position in the native wavelength units of the source dataset.

<i>All variables perturbed simultaneously</i>			
Tolerance level	1%	3%	5%
Peak jitter (std)	1.028	1.270	1.338
Peak spread (range)	2.964	3.000	3.000
<i>One variable at a time, 3% tolerance</i>			
Variable	Jitter (std)	Share of total	Rank
Λ	1.231	76%	1
d_2	0.244	15%	2
d_3	0.139	9%	3
d_1	0.008	< 1%	4

Shares are of the summed per-variable volatilities and do not account for interaction between variables, which is why they need not reproduce the simultaneous-perturbation jitter in the upper panel.

S6.3 Limitation of the peak-detection step

The peak spread in Table S5 is 2.964–3.000, against a wavelength grid that spans 3.000 units in total. A spread equal to the width of the search interval means the argmax is reaching the grid boundary in a substantial fraction of trials rather than tracking an interior resonance, and each such trial contributes a saturated rather than a physical peak position.

The consequence is that the jitter magnitudes in Table S5 are upper bounds. The variable ranking is more robust than the magnitudes, since boundary saturation affects all four variables under the same protocol, and the two-orders-of-magnitude separation between Λ and d_1 is far larger than the effect could plausibly manufacture. We report the numbers as published and flag the limitation; a peak finder that rejects boundary maxima, or a wider wavelength grid, would be needed before treating the magnitudes as calibrated.

References

- [1] Zelaci A, Yasli A, Kalyoncu C, et al (2021) Generative adversarial neural networks model of photonic crystal fiber based surface plasmon resonance sensor. *Journal of Lightwave Technology* 39(5):1515–1522. <https://doi.org/10.1109/JLT.2020.3035580>